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# SET B

## Department of Electrical Engineering

Indian Institute of Technology, Madras

**EE 2005 Electrical Machines - Quiz II**

Class: II B.Tech

Date: 24th March 2025

Duration: 50 min.

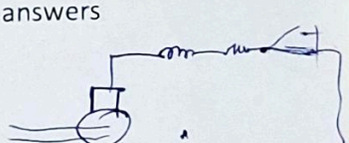
Max. Marks: 20

**Name:** V. Eswar Gupta **Roll no:** EE23B085

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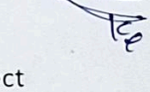
**Instructions to Candidates**

1. Answer ALL questions.
2. For all questions, fill in the blanks indicated with correct answers, or circle the correct option in the case of MCQs. There may be one or more correct options. Marks will be given only if all the correct option(s) is/ are marked correctly.
3. Do all intermediate numerical computations to 6 significant digits after decimal point in scientific notation.
4. All final answers when written down should be with 3 significant digits in scientific notation.
5. Mark your answers in this question paper itself and return it. You can use the reverse side of the sheets for any working needed. Do not write **ANYTHING** other than the final answers in the front sides.
6. Question number 1-12 carries 1 Marks each.



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$E_b \propto \phi\omega \rightarrow \omega \propto 1/\phi \propto 1/I_A$




1. Statement: "DC series motor cannot be operated at no load". What is the correct reason behind it?
   a) Above statement is false.
   b) It exhibits high starting torque.
   c) No load speed is much higher than the rated speed.
   d) No load speed is much lesser than the rated speed.

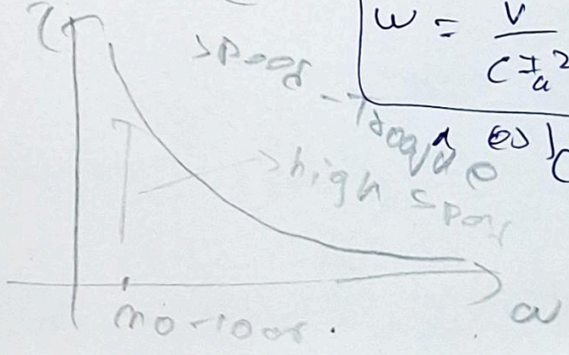
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$V = \omega K_e + I_a R$

$V = \omega C I_a^2 + I_a R$

$V - I_a R = \omega C I_a^2$

$\omega = \frac{V}{C I_a^2} - \frac{R}{C I_a}$ eq 1




correct

ans:- C

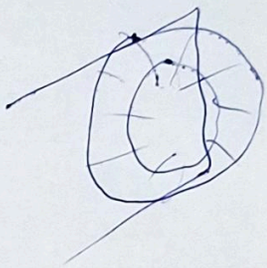
$I_a = \frac{V - E_b}{R_a}$

2. For regenerative braking to take place in a DC motor when it is rotating in anticlockwise direction, following condition(s) has/have to be true. (motor like generator)
   a) Back emf should be higher than the supply voltage.
   b) Back emf should be lesser than the supply voltage.
   c) Rotor should rotate in the clockwise direction.
   d) Current should flow from machine to the source.

3. Which statement(s) is/are true for stable operation of a DC machine.
   a) Generated torque (electromagnetic torque) and speed will be in the same direction for motoring operation.
   b) Load torque and speed will be in opposite direction for motoring operation. (load opposes motion)
   c) Generated torque (electromagnetic torque) and speed will be in the same direction for generating operation. (opposite direction)
   d) Load torque and speed will be in the same direction for motoring operation.

4. Ideally the angle between field flux and armature flux of a DC machine are at
   a) Quadrature 90°
   b) At 120°
   c) In line
   d) At 45°

$\tau \propto \phi_{field}\, \phi_{armature} \sin(\theta)$

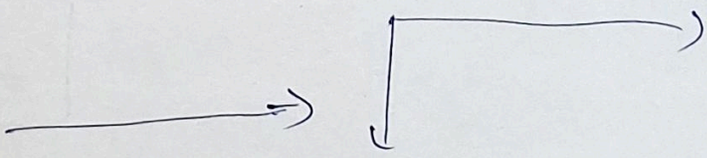


5. Which statement(s) is/are correct for a 3-Phase synchronous machine.
   a) AC supply is given to the rotor circuit.
   b) DC supply is given to the rotor circuit.
   c) 3-Phase balanced sinusoidally distributed windings are kept in the stator.
   d) At steady state rotor field and stator field runs at two different speeds with respect to the stator.

6. Which statement(s) is/are correct for a synchronous machine when it is operating at no load (Neither delivering real power nor reactive power) connected with an infinite bus?
   a) Per phase back emf (E) is lower in magnitude compared to per phase terminal voltage (V) but they are in the same phase.
   b) Per phase back emf (E) is higher in magnitude compared to per phase terminal voltage (V) but they are in same phase.
   c) Per phase back emf (E) is same in magnitude and phase as per phase terminal voltage (V).
   d) Per phase back emf (E) is same in magnitude as per phase terminal voltage (V) but quadrature in phase.

$\mathcal{E} = -\frac{d\phi}{dt}$



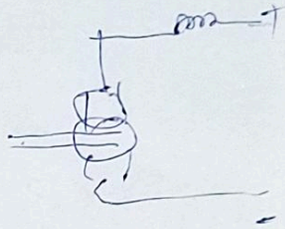


stator → almost non-zero core loss

7. which statement is **correct** in the context of core loss of an DC machine?
   a) Both Stator and Rotor exhibits similar amount of core loss.
   b) Stator and Rotor does not exhibit core loss as the flux flowing through them is DC.
   c) Rotor exhibits negligible core loss while stator exhibits finite core loss.
   d) Stator exhibits negligible core loss while rotor exhibits finite core loss.

8. Which part(s) of the DC machine is/are responsible for developing **unidirectional** torque in the armature?
   a) Stator stamping.
   b) Rotor lamination.
   c) Uniform air gap.
   d) Brush and commutator segments.

9. At no load condition, under excited synchronous machine connected to a grid behaves like a/an:
   a) Capacitor.
   b) Resistor.
   c) Transistor.
   d) Inductor.



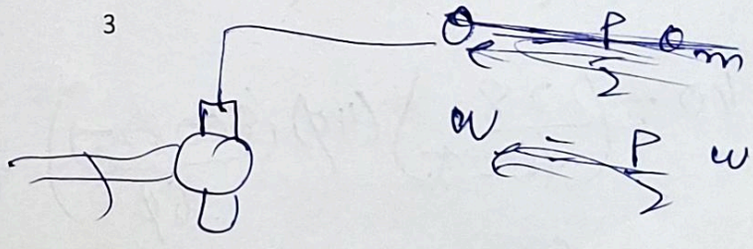
10. Statement: "Synchronous generator is delivering a finite electrical load (real power) at zero regulation to an infinite bus maintained at constant voltage and frequency". The current direction is from back emf to the grid voltage. What inference(s) certainly can be drawn from the above statement.
   a) Machine excitation is same as excitation at no load.
   b) Machine excitation is same as excitation at full load.
   c) Machine excitation is higher than excitation at no load.
   d) Machine is delivering leading power factor load.

11. The relative motion between stator field and rotor field at steady state of a synchronous machine is
   a) Higher than 100 rpm.
   b) Lower than 100 rpm.
   c) Zero rpm.
   d) Depends on the pole number of the machine.

12. A 20-pole synchronous machine is connected to a 60 Hz grid. The speed of the machine in RPM is

Ans. 360





$KI_a = 80 \quad K = 4$

13. The electromagnetic torque developed in a separately excited DC machine is 80 Nm for an armature current of 20A at 900 rpm speed.

a) Find the torque for a current of 30A when excitation is kept constant at same speed.

Torque value is ---------- 120 ---------- Nm. (2 Marks)

$240 = K(40)(125.6) + (40)(0.3) \Rightarrow K = 0.0454$

14. A 240 V DC series motor takes rated current of 40 A to drive rated load at 1200 rpm. The combined value of armature and series field resistance is 0.3 Ω.

a) Find the resistance must be added in series with the machine to get same torque at starting. The supply voltage is kept at 240 V. $240 = 40(R + 0.3) \Rightarrow R = 5.7$

Resistance value in Ohm..... 5.7 ..... (1 Marks)

b) Find the resistance to be added to obtain rated torque at 800 r.p.m. The supply voltage is kept at 240 V. $240 = 40(83.73)(0.0454) + 40(R + 0.3)$ $R = 1.898$

Resistance value to be added in Ohm..... 1.9 ..... (2 Marks)

15. A three- Phase star connected 50 Hz alternator has synchronous reactance of 0.5 Ohm. What is the excitation emf of the machine (Line to Line) such that it maintains 400 Volts at terminal when supplying 100 A of armature current.

a) at unity p.f. (1 Marks)

EMF induced (Line to Line) in Volts..... 409.2676386 ∠12.21634884

b) at 0.8 p.f. lagging. ∠8.71513 04

EMF induced (Line to Line) in Volts..... 457.24087 68 (2 Marks)

